

FINAL YEAR PROJECT • TM471

Smart Bahrain

A Mobile Platform for Reporting Traffic & Parking Violations

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Traffic & Parking Violations Are Hard to Track

Urban areas in Bahrain face daily congestion, blocked driveways, double parking, and reckless driving. Traditional monitoring relies on manual observation and CCTV — limited in coverage and slow to respond.

Most violations go unreported because citizens lack a direct, simple channel to inform authorities. The result: repeated offences, weak accountability and frustrated road users.



Limited Coverage

Manual monitoring can't watch every street.



Underreporting

No simple way for citizens to flag incidents.



Slow Response

Paper-based processes delay enforcement.

What Is Smart Bahrain?



A mobile-based reporting platform that lets citizens submit traffic and parking violations with photo, video and GPS evidence — and lets authorities review and act on them through a secure digital dashboard.



Citizen Mobile App

Report, attach evidence, track status, earn rewards.



Admin Dashboard

Review, assign, resolve or reject reports digitally.



Vision 2030 Aligned

Supports digital transformation and smart-city goals.

Aim & Objectives

AIM

Develop a mobile platform that lets citizens of Bahrain report traffic and parking violations quickly and efficiently — improving communication with authorities and supporting Bahrain's digital transformation.

SPECIFIC OBJECTIVES

- 1 Provide an easy-to-use mobile platform for real-time reporting
- 2 Enable authorities to receive and manage reports digitally
- 3 Improve road safety by reducing unreported violations
- 4 Encourage citizen participation through a reward system
- 5 Contribute to Bahrain's smart-city vision

Key Features for Citizens



CPR Login & Signup

Secure account creation with bcrypt-hashed passwords and JWT tokens.



Report Submission

Category, description, photo / video evidence and GPS location.



GPS & Map Selection

Auto-capture coordinates with map confirmation via React Native Maps.



My Reports & Tracking

View, search and filter reports by status: pending, under review, resolved or rejected.



SMS Notifications

Twilio-powered alerts on submission and status changes.



BHD Reward Wallet

Earn rewards for verified reports and request withdrawals.

Tools for Authorities & Moderators



Live Statistics

Total, pending, resolved and rejected reports plus monthly trends via Recharts.



Report Review

Inspect evidence, assign cases, resolve or reject with admin notes.



User Management

Search by name, CPR or phone; reset passwords securely.



Moderator Roles

Create, edit, delete moderators with email-based access control.



Role-Based Access

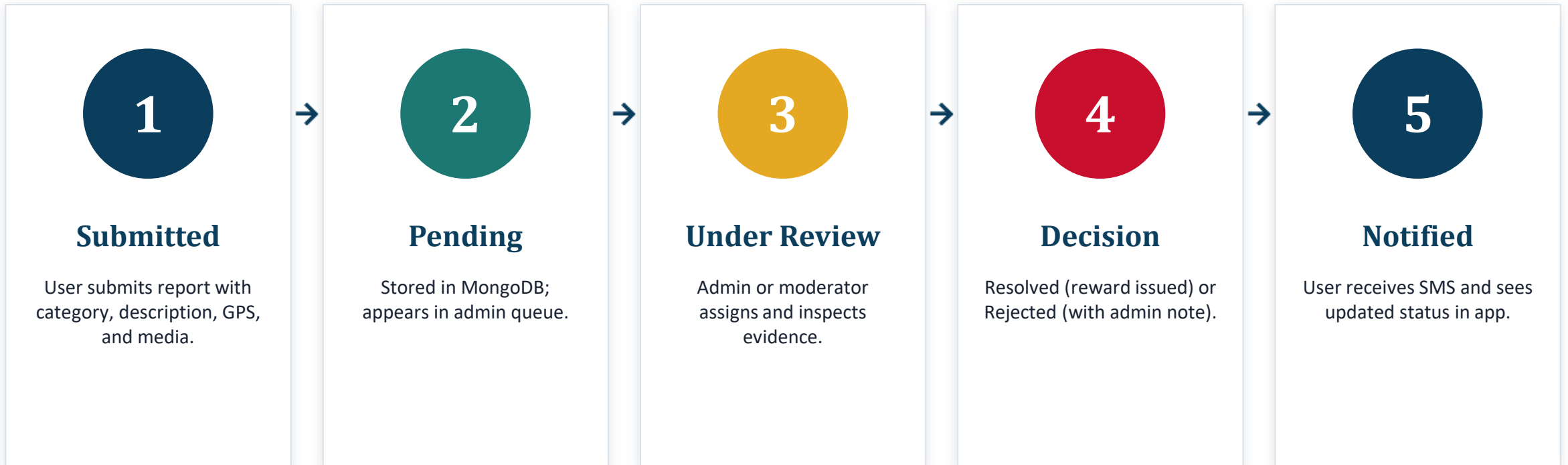
JWT middleware enforces admin / moderator / user permissions.



Resolved & Rejected Logs

Searchable archive of every decision with full audit visibility.

How a Report Flows Through the System



Every status change flows through the backend API — keeping the mobile app and admin dashboard perfectly in sync.

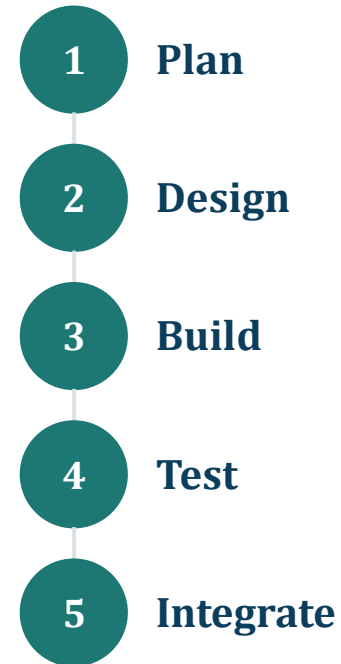
Incremental Development Model

WHY INCREMENTAL?

The system is naturally modular — auth, reporting, media upload, status tracking, dashboard. Each module was built, tested and integrated separately, with continuous supervisor feedback at every stage.

- ✓ Early testing & validation per increment
- ✓ Reduced project risk — issues stay local
- ✓ Continuous supervisor feedback loop
- ✓ Core features delivered & demoed early
- ✓ Easier debugging and future scaling

INCREMENT FLOW



Waterfall was rejected — too rigid for iterative features like media uploads & notifications.

How Smart Bahrain Compares

Feature	Traffic Prahari (India)	Saudi Balady	Smart Bahrain
Primary Focus	Traffic / parking	Municipal violations	Traffic / parking
Coverage Area	Delhi only	Saudi Arabia	Bahrain
Reward Model	Top contributors only	Partial fine for verified	Per verified report
Status Updates	Limited	Limited	Real-time
User Registration	General users	Certified Observer	Standard signup
Reporting Scope	Wide traffic offences	Broad municipal	Traffic & parking-specific

Smart Bahrain fills a clear gap — a Bahrain-specific, traffic-focused reporting tool with real-time tracking and per-report rewards.

BHD Reward System



REWARD ENGINE

BHD

Per-Verified-Report

When an admin marks a report Resolved, the backend calculates a reward based on the violation category and credits the user's wallet automatically.

HOW IT WORKS

Submit & Resolve

User reports a verified violation; admin approves it.

Auto-Calculation

Backend computes BHD reward by violation category.

Wallet & History

Balance, earning history and withdrawal status visible in-app.

Withdrawal Request

Validation against minimum amount and available balance.

Goals Achieved



Secure user signup, login & CPR-based authentication



Full report submission with media, GPS & category



Personal report tracking with search, filter & details



Admin dashboard with stats, review, assign & resolve



BHD reward engine with withdrawal validation



Role-based access control (user / moderator / admin)



Exceeded scope:

SMS notifications, emergency contacts with QR, dashboard analytics, moderator management.

Future Work



Push Notifications

Instant in-app alerts for status, rewards and withdrawals.



Arabic Language Support

Bilingual interface for wider local accessibility.



Offline Submissions

Cache reports locally and sync when connection returns.



Bank Integration

Automatic withdrawal payouts to user accounts.



AI Verification

Image / video analysis to filter false reports.



Heatmap & Live Map

Real-time incident map and traffic violation heatmap.

CONCLUSION

A practical foundation for digital traffic reporting.

Smart Bahrain combines mobile reporting, GPS, evidence handling, role-based dashboards and rewards into one connected system — a measurable contribution to Bahrain's smart-city vision.

Thank You • Questions?